Capacitively-Coupled Chopper Instrumentation Amplifier for Linear Hall Effect Sensor

IEEE SSCS Chipathon 2026

Team Chop

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Contents

1 Project Information

Project Name	CCIA for Linear Hall Effect Sensor
Team Members	alang
Review Date	
Review Version	
Track	B

[1]

2 Design Objective

The aim of this Chipathon project is to design and characterize a monolithic magnetic sensing system based on the Hall Effect. If tapeout is completed, the specifications of the sensor and readout circuit will be characterized individually.

The target specifications are split into three parts: (1) Hall Effect sensor, (2) readout circuit, (3) full magnetic sensing system. See the Appendix for the full specifications documents for both components. Below are three tables summarizing the desired specifications for the three parts.

(insert sensor specifications table)

(insert readout circuit specifications table)

(insert full system specifications)

The motivation for this project in the context of the Chipathon project is to provide reusable IP to the open-source silicon community. Priority will be given to designing a simple system that can be well characterized as opposed to a novel system. Overall, if the tapeout is successful, future designers would be able to reuse and customize both the sensor design and the readout circuit for their specific application. In any case, future Chipathon participants will be able to learn from the successes and failures of the project.

3 System Overview

A block diagram of the system is shown in Fig. 1.

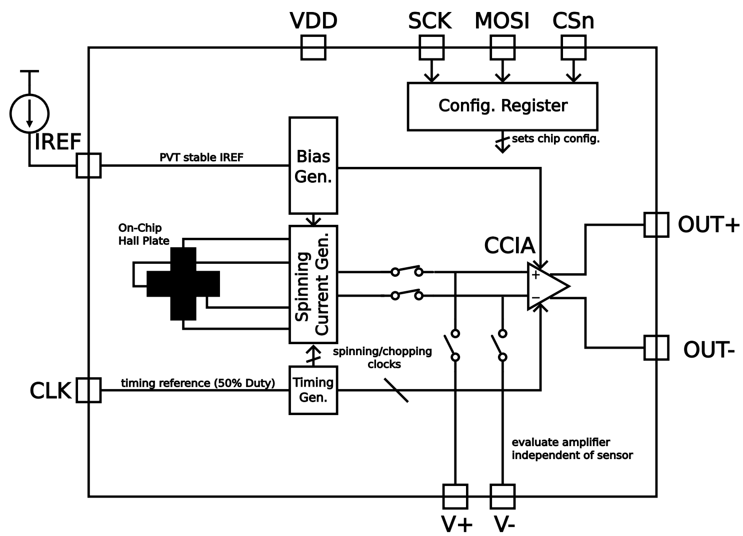

Figure 1: Block diagram of the proposed architecture.

In the diagram, an on-chip horizontal Hall plate converts an incoming magnetic field into a small voltage that is amplified by the instrumentation amplifier prior to being driven to the DUT output pins. The entire design operates with a single supply. Aside from the SPI configuration interface (only used during idle periods), the entire design operates in a single synchronous clock domain derived from the external clock signal (CLK). Two additional pins are used as analog debug pins (V+ and V-). Based on the configuration, these pins will be used to characterize the sensor and the readout circuit individually.

A full pinout table is given in Table 1. The remaining subsections discuss the detailed operation of each block.

3.1 On-Chip Hall Plate

3.2 Bias Generation

3.3 Spinning Current Generation

3.4 Timing Signal Generation

3.5 Configuration Register

3.6 Instrumentation Amplifier

Table 1: Chip pinout description.

Analog Pins		
IREF	Analog	Current reference
OUT+	Analog	Differential output positive terminal
OUT-	Analog	Differential output negative terminal
VTP+	Analog	Positive analog test point (differential probe node)
VTP-	Analog	Negative analog test point (differential probe node)
Digital Pins		
SCK	Digital Input	SPI configuration clock
MOSI	Digital Input	SPI configuration data input
CSn	Digital Input	SPI chip select
CLK	Digital Input	Chopping/spinning clock reference
Supply Pins		
VDD	Power	Supply input
VSS	Power	Ground (shared with other Chipathon blocks)

4 Schematic Summary

4.1 Sheet 1.sch

- Purpose
- Inputs and Outputs
- Important Design Decisions
- Interfaces to other blocks

4.2 Sheet 2.sch

- Purpose
- Inputs and Outputs
- Important Design Decisions
- Interfaces to other blocks

5 Design Assumptions

As stated in the specifications, the following assumptions are made within the design.

- The circuit shall be designed to operate from a single 3.3V supply
- The circuit shall be designed to operate over the temperature range of ($-40\,^{\circ}$C - $125\,^{\circ}$C)
- The specifications of each circuit shall be simulated over all worst-case process corners
- Each analog output (both (+) and (-)) of the readout circuit shall be capable of driving a 10 pF off-chip load
- To test the full system, an external magnetic field will be applied via a Helmholtz coil pair. A printed circuit board will be designed to supply all DUT inputs and condition the differential readout circuit output prior to digitization.

6 Verification Status

6.1 Functional Simulations

6.2 Corner Simulations

6.3 Monte Carlo

6.4 Sign-off

- ERC Status
- DRC Status
- LVS Status

7 Design Checklist

7.1 Functionality

7.2 Analog

7.3 Digital

7.4 Mixed-Signal

7.5 Reliability

7.6 Documentation

8 Open Issues

List of open issues with:

- Description
- Impact
- Proposed Solution
- Owner

9 Questions for Reviewers

10 Review Outcome

Appendix

References

[1] L. Lamport, *LaTeX: A Document Preparation System*, 2nd ed. Addison-Wesley, 1994.